

POSTERIOR PROBABILITIES FOR SOME REGRESSION HYPOTHESES

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Summary: Posterior probabilities are derived for a number of standard hypotheses in linear regression. Two classes of prior distributions are considered. One class is a generalization of the proposal by Zellner and Siow (1980), and the other the conjugate class for the normal model. Lower bounds on the Bayes factor over these classes are derived. Comparisons with the p -value of classical tests show the same discrepancies found by a number of authors in different settings. Model selection in the framework of hypothesis testing is also considered.

1. INTRODUCTION

The general model for the regression analysis is

$$(1.1) \quad y = \alpha \underline{e} + X\beta + \epsilon,$$

where $\underline{e}$ is an $n \times 1$ vector of ones, α an unknown scalar, X an $n \times k$ matrix of rank k , β a $k \times 1$ vector of unknown regression coefficients, and ϵ is an $n \times 1$ vector of error terms. It is assumed that elements of ϵ are independently drawn from a normal population with mean zero and unknown finite variance σ^2 .

A number of authors have contributed to Bayesian Regression Analysis and amongst them are Zellner and Siow (1980), Zellner (1984), Box and Hill (1967), Dickey (1971, 1974), Lempers (1971), Gaver and Geisel (1974), Atkinson (1978), Geisser and Eddy (1979), Smith and Spiegelhalter (1980), Albert (1988), Mitchell and Beauchamp (1988), Spiegelhalter and Smith (1982), Lindley and Smith (1972) and Rao and Wu (1989). Most of the cited contributions, however, concentrate on model selection.

In this paper two different prior distributions will be considered. One is a normal prior on the unknown regression coefficients with vague priors on the nuisance parameters. The other approach is a modified version of the prior suggested by Zellner and Siow (1980).

Zellner and Siow (1980) give an approach which is an extension of that of Jeffreys (1961). For the general model as given above, when testing two hypotheses they place equal prior probabilities of $\frac{1}{2}$ on both hypotheses as is done in this paper.

In testing the hypotheses

$$H_0: \underline{\beta} = 0, \quad -\infty < \alpha < \infty \quad \text{and} \quad 0 < \sigma < \infty$$

$$H_1: \underline{\beta} \neq 0, \quad -\infty < \alpha < \infty \quad \text{and} \quad 0 < \sigma < \infty,$$

under both hypotheses H_0 and H_1 they employ a noninformative prior distribution for α and σ , that is

$$(1.2) \quad p(\alpha, \sigma | H_0) \propto 1/\sigma \quad -\infty < \alpha < \infty \quad \text{and} \quad 0 < \sigma < \infty.$$

Under H_1 they assume that

$$p(\alpha, \underline{\beta}, \sigma | H_1) \propto f(\underline{\beta} | \sigma) / \sigma \quad -\infty < \alpha < \infty \quad \text{and} \quad 0 < \sigma < \infty$$

where

$$f(\underline{\beta} | \sigma) = c |X'X/n\sigma^2|^{\frac{1}{2}} \left[1 + \frac{\underline{\beta}' X' X \underline{\beta}}{n\sigma^2} \right]^{-\frac{(k+1)}{2}} \quad -\infty < \beta_i < \infty$$

and

$$(1.3) \quad c = \Gamma[(k+1)/2] / \pi^{\frac{(k+1)}{2}}.$$

In their subsequent analysis they established results which, as they put it in their concluding remarks, are in line with Jeffreys', Lindley's and some others' previous results and that they have found that sampling theorists' usual 0,05 critical value of tests statistics can be far from a Bayesian posterior odds indifference value of one under a variety of circumstances.

2. GENERAL THEORY

Suppose X having density $f(x|\theta)$ is observed, with θ being an unknown element of the parameter space $\Omega = \{\Omega_0, \Omega_1, \dots, \Omega_k\}$ and that it is desired to test:

$$H_0 : \theta \in \Omega_0$$

against the multiple alternatives

$$(2.1) \quad \begin{array}{l} H_1 : \theta \in \Omega_1 \\ \vdots \\ H_k : \theta \in \Omega_k \end{array}$$

Let the prior density on θ be

$$(2.2) \quad p(\theta) = \begin{cases} p_0(\theta), & \theta \in \Omega_0 \\ \vdots \\ p_k(\theta), & \theta \in \Omega_k \end{cases}$$

with respective prior probabilities $\pi_0, \dots, \pi_k$.

The posterior density of θ given $X = x$ will be given by

$$p(\theta|x) = \begin{cases} p_0(\theta|x), & \theta \in \Omega_0 \\ \vdots \\ p_k(\theta|x), & \theta \in \Omega_k \end{cases},$$

where

$$(2.3) \quad \lambda_i p_i(\theta|x) = \frac{\pi_i p_i(\theta) f(x|\theta \in \Omega_i)}{\sum_{j=0}^k \pi_j \int_{\Omega_j} p_j(\theta) f(x|\theta \in \Omega_j) d\theta}, \quad \theta \in \Omega_i,$$

and λ_i is the posterior probability of H_i for $i = 0, \dots, k$. When $k = 1$, the Bayes factor in favour of H_0 is defined as

$$(2.4) \quad BF = \int_{\Omega_0} p_0(\theta) f(x|\theta \in \Omega_0) d\theta / \int_{\Omega_1} p_1(\theta) f(x|\theta \in \Omega_1) d\theta,$$

which is the weighted likelihood ratio of H_0 to H_1 .

The posterior probability of H_0 can then be shown to be

$$(2.5) \quad \lambda_0 = \left[1 + \frac{(1 - \pi_0)}{\pi_0} (BF)^{-1} \right]^{-1}.$$

This λ_0 is used as a measure of evidence in favour of the null hypothesis.

From (2.4) it can be seen that the Bayes factor is independent of the subjective probability π_0 and because of this some statisticians prefer it to the posterior probability as a measure of evidence for or against the null hypothesis.

In the determination of the posterior probability for a hypothesis H_i , there is need to specify π_i , $i = 1, \dots, k$, which is the prior for H_i given k hypotheses. In this paper $\pi_i = \frac{1}{k}$ will be used, this being interpreted as an objective and reasonable choice (see Berger and Sellke (1987)) for the case $k = 2$.

3. TESTING FOR SLOPE IN MULTIPLE REGRESSION

In this section the procedure will be derived for multiple regression and the straight line will follow as a special case.

We consider model (1.1) and assume that:

1. the variables in X are measured in terms of deviations from their respective sample means;
2. elements of ϵ are independently drawn from a normal population with mean zero and unknown finite variance σ^2 .

The following hypotheses will be of interest:

$$(3.1) \quad \text{against } H_0: \underline{\beta} = \underline{0}, \quad -\infty < \alpha < \infty, \quad 0 < \sigma^2 < \infty$$

$$H_1: \underline{\beta} \neq \underline{0}, \quad -\infty < \alpha < \infty, \quad 0 < \sigma^2 < \infty.$$

Two different approaches which differ in the prior distributions for the unknown parameters will be considered.

3.1 Let

$$p(\underline{y} | \alpha, \sigma^2, H_0) = \left[\frac{1}{2\pi\sigma^2} \right]^{\frac{n}{2}} \exp \left[-\{ \nu_1 s_1^2 + n(\alpha - \bar{y})^2 \} / 2\sigma^2 \right]$$

and

$$p(\underline{y} | \alpha, \underline{\beta}, \sigma^2, H_1) = \left[\frac{1}{2\pi\sigma^2} \right]^{\frac{n}{2}} \exp \left\{ - \left[\nu_2 s_2^2 + n(\alpha - \bar{y})^2 + (\underline{\beta} - \hat{\underline{\beta}})' X' X (\underline{\beta} - \hat{\underline{\beta}}) \right] / 2\sigma^2 \right\}$$

where

$$\nu_1 = n - 1, \quad \nu_2 = n - k - 1, \quad \nu_1 s_1^2 = \Sigma (y_i - \bar{y})^2,$$

$$\nu_2 s_2^2 = (\underline{y} - \bar{y}\underline{e} - X\hat{\underline{\beta}})' (\underline{y} - \bar{y}\underline{e} - X\hat{\underline{\beta}}) \quad \text{and} \quad \hat{\underline{\beta}} = (X'X)^{-1} X' \underline{y}.$$

Suppose that under H_0 , $p(\alpha, \sigma^2 | H_0) \propto \frac{1}{\sigma^2}$ and under H_1

$$p(\alpha, \underline{\beta}, \sigma^2) \propto f(\underline{\beta} | \sigma^2) / \sigma^2$$

with $f(\underline{\beta} | \sigma^2) = c |X'X / n\gamma\sigma^2|^{\frac{1}{2}} / (1 + \underline{\beta}' X' X \underline{\beta} / n\gamma\sigma^2)^{\frac{k+1}{2}}$

where $c = \Gamma\left(\frac{k+1}{2}\right) / \pi^{\frac{(k+1)}{2}}$ and γ is an unknown positive real number.

This is a Cauchy type prior as was suggested by Zellner and Siow (1980), but with $\gamma = 1$.

The numerator of the Bayes factor will thus be

$$\begin{aligned}
 (3.1.1) \quad I_N &= \int \int \left[\frac{1}{2\pi\sigma^2} \right]^{\frac{n}{2}} \frac{1}{\sigma^2} \exp \left\{ - \left[\nu_1 s_1^2 + n(\alpha - \bar{y})^2 \right] / 2\sigma^2 \right\} d\alpha d\sigma^2 \\
 &= \left[\frac{1}{2\pi} \right]^{\frac{n-1}{2}} \left[\frac{1}{n} \right]^{\frac{1}{2}} \left[\frac{2}{\nu_1 s_1^2} \right]^{\frac{\nu_1}{2}} \Gamma \left[\frac{\nu_1}{2} \right].
 \end{aligned}$$

The denominator is as follows:

$$\begin{aligned}
 I_D &= \int \dots \int \left[\frac{1}{2\pi\sigma^2} \right]^{\frac{n}{2}} \frac{1}{\sigma^2} f(\underline{\beta} | \gamma\sigma^2, H_1) \\
 &\times \exp \left\{ - \left[\nu_2 s_2^2 + n(\alpha - \bar{y})^2 + (\underline{\beta} - \hat{\underline{\beta}})' X' X (\underline{\beta} - \hat{\underline{\beta}}) \right] / 2\sigma^2 \right\} d\alpha d\underline{\beta} d\sigma^2.
 \end{aligned}$$

The approximate integration with respect to α , the elements of $\underline{\beta}$ and σ^2 (Zellner and Siow (1980 page 592)) yields

$$(3.1.2) \quad I_D = \left[\frac{1}{2\pi} \right]^{\frac{n-1}{2}} \left[\frac{1}{n} \right]^{\frac{1}{2}} \left[\frac{2\pi}{\nu_2} \right]^{\frac{k}{2}} \Gamma \left[\frac{\nu_1}{2} \right] \left[\frac{2}{\nu_2 s_2^2} \right]^{\frac{\nu_1}{2}} c / (1 + \hat{\underline{\beta}}' X' X \hat{\underline{\beta}} / \gamma \nu_2 s_2^2)^{\frac{(k+1)}{2}}.$$

Thus

$$\begin{aligned}
 (3.1.3) \quad BF &= \frac{1}{c} \left[\frac{\gamma \nu_2}{2\pi} \right]^{\frac{k}{2}} \left[\frac{\nu_2 s_2^2}{\nu_1 s_1^2} \right]^{\frac{\nu_1}{2}} (1 + \hat{\underline{\beta}}' X' X \hat{\underline{\beta}} / \gamma \nu_2 s_2^2)^{\frac{(k+1)}{2}} \\
 &= (\pi)^{\frac{1}{2}} \left[\frac{\gamma \nu_2}{2} \right]^{\frac{k}{2}} \left[\Gamma \left[\frac{k+1}{2} \right] \right]^{-1} \left[1 + \frac{kF}{\nu_2} \right]^{-\frac{\nu_1}{2}} \left[1 + \frac{kF}{\gamma \nu_2} \right]^{\frac{(k+1)}{2}}
 \end{aligned}$$

where $F = \frac{\hat{\underline{\beta}}' X' X \hat{\underline{\beta}}}{k s_2^2}$ is the usual classical F -statistic for testing the hypothesis in (3.1.2) with k and ν_2 degrees of freedom.

The posterior probability of H_0 will then be

$$(3.1.4) \quad \lambda = (1 + (BF)^{-1})^{-1}.$$

It thus follows that the posterior probability is a function of γ . Since there is no default value of γ and the statistician may not have enough information to specify its magnitude, it is helpful here to look at the behaviour of this posterior probability as a function of γ . Figure 3.1 shows the graph of the posterior probability of H_0 as a function of γ for a given p -value.

One can obtain the minimum of BF and hence of λ with respect to γ . It can be shown that this minimum will occur at

$$(3.1.5) \quad \hat{\gamma} = \frac{F}{\nu_2}$$

and this will give the minimum Bayes factor as

$$(3.1.6) \quad BF^* = (\pi)^{\frac{1}{2}} \left[\Gamma \left[\frac{k+1}{2} \right] \right]^{-1} \left[\frac{F}{2} \right]^{\frac{k}{2}} \left[1 + \frac{kF}{\nu_2} \right]^{\frac{\nu_1}{2}} (1+k)^{\left(\frac{k+1}{2} \right)}$$

and this leads to the minimum of the posterior probability λ^* .

It is seen from (3.1.6) that for fixed F , the minimum posterior probability is a function of the sample size n , and it is possible to establish the limit as $n \rightarrow \infty$ since

$$(3.1.7) \quad \lim_{n \rightarrow \infty} BF^* = BF^{**} = (\pi)^{\frac{1}{2}} \left[\Gamma \left[\frac{k+1}{2} \right] \right]^{-1} \left[\frac{F}{2} \right]^{\frac{k}{2}} (1+k)^{\left(\frac{k+1}{2} \right)} e^{-\frac{kF}{2}}$$

Table 3.1 gives the minimum posterior probabilities for H_0 as well as the posterior probabilities for H_0 using the approach of Zellner and Siow, that is with $\gamma = 1$, for selected values of n and k with F corresponding to the critical value at the 5% level of significance.

For each combination of n and k the first row gives the lower bound on the posterior probability of H_0 whereas the second row gives the posterior probability of H_0 when $\gamma = 1$.

TABLE 3.1: Lower bounds on posterior probabilities of H_0

k	n	5	10	18	32	∞
3		0,0703	0,3028	0,3830	0,4190	0,4586
		0,3852	0,3051	0,4948	0,6538	1
5			0,1556	0,3311	0,4107	0,4680
			0,1606	0,4481	0,7763	1
9				0,1116	0,3105	0,5421
				0,1340	0,7447	1

3.2 In this second approach assume that under both H_0 and H_1 $\alpha \sim N(m, \gamma_1 \sigma^2)$, $h(m, \sigma^2) \propto \frac{1}{\sigma^2}$ and under H_1 , $\beta \sim N(0, \gamma \sigma^2)$.

The numerator of the Bayes factor will thus be

$$\begin{aligned}
 (3.2.1) \quad I_N &= (2\pi)^{-\frac{n}{2}} \int \dots \int \left[\frac{1}{\sigma^2} \right]^{\frac{n}{2}+1} \exp[-\nu_1 S_1^2 + n(\alpha - \bar{y})^2 / 2\sigma^2] \\
 &\quad \times (2\pi\gamma_1\sigma^2)^{-\frac{1}{2}} \exp\left[-\frac{(\alpha - m)^2}{2\gamma_1\sigma^2}\right] dmd\alpha d\sigma^2 \\
 &= n^{-\frac{1}{2}} (2\pi)^{-\left(\frac{n-1}{2}\right)} \Gamma\left[\frac{n-1}{2}\right] \left[\frac{2}{\nu_1 S_1^2}\right]^{\frac{n-1}{2}}
 \end{aligned}$$

The denominator is given by

$$\begin{aligned}
 I_D &= (2\pi)^{-\frac{n}{2}} \int \dots \int \left[\frac{1}{\sigma^2} \right]^{\frac{n+1}{2}} (2\pi\gamma_1\sigma^2)^{-\frac{1}{2}} (2\pi\gamma\sigma^2)^{-\frac{k}{2}} \exp \left[-\frac{(\alpha-m)^2}{2\gamma_1\sigma^2} - \frac{\beta'\beta}{2\gamma\sigma^2} \right] \\
 (3.2.2) \quad &\times \exp \left[-[\nu_2 s_2^2 + n(\alpha - \bar{y})^2 + (\beta - \hat{\beta})' X' X (\beta - \hat{\beta}) / 2\sigma^2] \right] d\alpha d\beta d\sigma^2 \\
 &= n^{-\frac{1}{2}} (2\pi)^{-\frac{(n-1)}{2}} \Gamma \left[\frac{n-1}{2} \right] |\gamma X' X + I|^{-\frac{1}{2}} \\
 &\quad \times \left[\frac{2}{\nu_2 s_2^2 - \gamma \hat{\beta}' X' X (\gamma X' X + I)^{-1} X' X \hat{\beta} + \hat{\beta}' X' X \hat{\beta}} \right]^{\frac{n-1}{2}}
 \end{aligned}$$

so that the Bayes factor is

$$(3.2.3) \quad BF = \left[1 - \frac{\gamma \hat{\beta}' X' X (\gamma X' X + I)^{-1} X' X \hat{\beta}}{\nu_1 S_1^2} \right]^{\frac{\nu_1}{2}} |\gamma X' X + I|^{\frac{1}{2}}$$

since $\nu_1 S_1^2 = \nu_2 s_2^2 + \hat{\beta}' X' X \hat{\beta}$.

It is noticed again that the Bayes factor is a function of the data and the unknown γ , and, unlike section 3.1, this BF cannot be written as a function of the classical test statistic F .

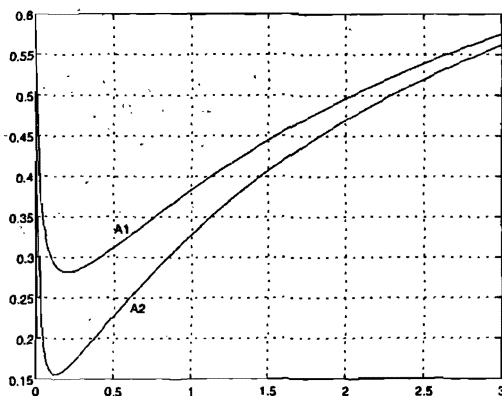


FIGURE 3.1: Posterior probabilities for the two approaches as function of γ

Figure 3.1 also shows the posterior probability of H_0 as a function of γ for $n = 25$, $k = 2$ and $F = 4.4476$, which is just significant at $\alpha = 0.025$. In this figure A_1 and A_2 refer to the approaches treated in sections 3.1 and 3.2 respectively.

It is seen from the graphs that the first approach results in higher posterior probabilities for H_0 and the case $\gamma = 1$, which would result in the prior as used by Zellner and Siow, gives even higher posterior probability for H_0 . It is, however, interesting to note that for the second approach, $\lambda = 0.5$ when $\gamma = 0$, which implies the data is ignored and for both approaches $\lambda \rightarrow 1$ as $\gamma \rightarrow \infty$ and Lindley's paradox occurs (Lindley, 1957).

These results conform to the results obtained in the past since the lower bound of posterior probabilities for any of the two approaches exceeds the corresponding p -value of 0.025.

4. TESTING FOR SLOPE IN A STRAIGHT LINE

If the regression model given by (1.1) is considered with $k = 1$ then it becomes

$$(4.1) \quad y_i = \alpha + \beta x_i + \varepsilon_i, \quad i = 1, \dots, n \text{ with } E(\varepsilon_i) = 0, \quad V(\varepsilon_i) = \sigma^2,$$

4.1 For the prior from section 3.1, the Bayes factor for testing the hypotheses (3.1) follows for the special case as

$$(4.1.1) \quad BF = \left[\frac{n\gamma\pi}{2} \right]^{\frac{1}{2}} \left[1 + \frac{\hat{\beta}^2 \Sigma x_i^2}{\gamma \nu_2 s_2^2} \left(\frac{\nu_2 s_2^2}{\nu_1 s_1^2} \right)^{\frac{n-1}{2}} \right]$$

where $\nu_2 = n - 2$, $\nu_2 s_2^2 = \Sigma (y_i - \bar{y} - \hat{\beta} x_i)^2$, and $\hat{\beta} = \frac{\Sigma x_i (y_i - \bar{y})}{\Sigma x_i^2}$.

If one writes $F = \frac{\hat{\beta}^2 \Sigma x_i^2}{s_2^2}$ which is the usual F statistic with 1 and ν_2

degrees of freedom and uses the relationship $\nu_2 s_2^2 + \hat{\beta}^2 \Sigma x_i^2 = \nu_1 s_1^2$, then

$$(4.1.2) \quad BF = \left[\frac{\nu_2 \gamma \pi}{2} \right]^{\frac{1}{2}} \left[1 + \frac{F}{\gamma \nu_2} \right] \left[1 + \frac{F}{\nu_2} \right]^{-\frac{\nu_1}{2}},$$

as in Zellner and Siow (1980) page 602. The posterior probability of H_0 follows from (3.1.4) under the assumption that $\pi_0 = \frac{1}{2}$.

Figure 4.1 shows the posterior probability of the null hypothesis as a function of γ . The lower bound with respect to γ is given by

$$(4.1.3) \quad BF^* = 2 \left[\frac{\pi F}{2} \right]^{\frac{1}{2}} \left[1 + \frac{F}{\nu_2} \right]^{-\frac{\nu_1}{2}}.$$

Also,

$$(4.1.4) \quad \lim_{n \rightarrow \infty} BF^* = BF^{**} = 2 \left[\frac{\pi F}{2} \right]^{\frac{1}{2}} e^{-\frac{F}{2}}$$

and λ^{**} follows.

Table 4.2 gives the lower bound on the posterior probabilities for H_0 as well as the posterior probabilities for H_0 using Zellner and Siow prior, that is $\gamma = 1$, for selected values of n with F corresponding to the critical value at the 5% level of significance.

TABLE 4.1: Posterior probabilities of H_0

	n				
	4	10	17	32	∞
λ^*	0,2471	0,3684	0,3915	0,4049	0,4186
$\lambda \gamma = 1$	0,3681	0,3732	0,4324	0,5096	1

The minimum posterior probability of H_0 is much higher than the classical p -value for the same data, indicating the readiness of classical statistics to reject the null hypothesis even though the data does not provide substantial evidence against it.

4.2 In the second approach with priors as in 3.2, the Bayes factor is

$$BF = (1 + \gamma \Sigma(x_i - \bar{x})^2)^{\frac{1}{2}} \left[1 - \left[\frac{\gamma \Sigma(x_i - \bar{x})^2 + 1}{\gamma r^2 \Sigma(x_i - \bar{x})^2} \right]^{-1} \right]^{\frac{n-1}{2}}$$

where r is the sample Pearson's correlation coefficient. Since $F = \frac{r^2(n-2)}{1-r^2}$, where F is the classical test statistic for testing the hypotheses in (3.1), one can write

$$(4.2.1) \quad BF = (1 + \gamma \Sigma(x_i - \bar{x})^2)^{\frac{1}{2}} \times \left[1 - \left\{ ((n-2)F^{-1} + 1) \left[1 + \frac{1}{\gamma \Sigma(x_i - \bar{x})^2} \right] \right\}^{-1} \right]^{\frac{n-1}{2}}$$

It thus follows that λ is a function of the classical test statistic F , n , $\Sigma(x_i - \bar{x})^2$ and the unknown γ . Figure 4.1 gives the graph of λ as a function of γ .

Further for any given data, the lower bound of the Bayes factor as well as that of the posterior probability of H_0 with respect to γ is attained at

$$(4.2.2) \quad \hat{\gamma} = \frac{F-1}{\Sigma(x_i - \bar{x})^2}, \quad F > 1$$

and at $\hat{\gamma} = 0$ for $F \leq 1$, which will be $\pi_0 = 0.5$.

This lower bound is

$$(4.2.3) \quad BF^* = F^{\frac{1}{2}} \left[\frac{n-1}{n-2+F} \right]^{\frac{n-1}{2}}$$

and λ^* follows from (3.1.4).

Also,

$$(4.2.4) \quad \lim_{n \rightarrow \infty} BF^* = BF^{**} = F^{\frac{1}{2}} e^{-\frac{(F-1)}{2}}$$

Table 4.2 gives the minimum posterior probabilities for H_0 for selected values of n with F corresponding to the critical value at the 5% level of significance.

TABLE 4.2: Lower bounds on the posterior probabilities of H_0

	n				
	4	10	17	32	∞
λ^*	0,2406	0,2833	0,3008	0,3110	0,3213

From table 4.2 it is seen that the posterior probabilities are less than those obtained in the first approach although again these posterior probabilities of H_0 are still greater than the classical p -value for the same data.

As a further comparison of the two approaches figure 4.1 gives the

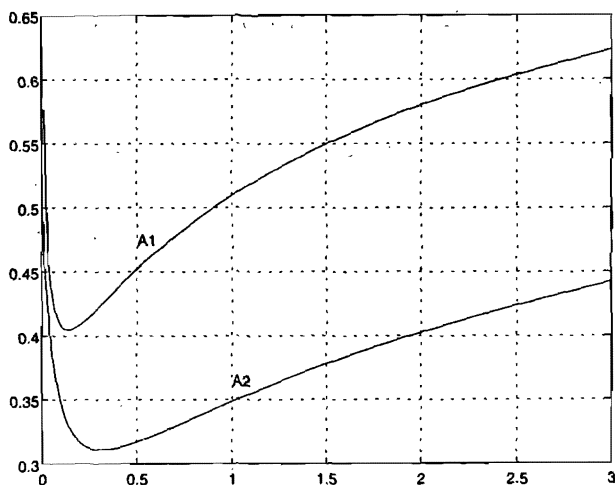


FIGURE 4.1: Graphs of the posterior probability of H_0 for the two approaches

graphs of the posterior probabilities of H_0 for both approaches as a function of γ . For this graph $\Sigma(x_i - \bar{x})^2 = 10$, $n = 32$, and $F = 4.171$ which is the classical critical value for testing the hypotheses in (3.1) at the 5% level of significance. A_1 and A_2 are the approaches treated in sections 4.1 and 4.2 respectively.

As expected, the posterior probability for the heavier tailed Cauchy prior is higher than for the normal prior. Also, it should be noted that the parameter γ does not play exactly the same role in the two approaches.

5. TESTING FOR PARALLELISM OF TWO STRAIGHT LINES

Given the regression model $y_{ij} = \alpha_i + \beta_i x_{ij} + \varepsilon_i$ with $E(\varepsilon_i) = 0$, $\text{var}(\varepsilon_i) = \sigma^2$ and $y_i \sim N(\alpha_i + \beta_i x_i, \sigma^2)$ for $i = 1, 2$, it is the purpose of this section to derive a Bayesian test of the hypothesis that the two lines are parallel and thus the hypotheses will be formulated as

$$H_0 : \beta_1 = \beta_2 = \beta_0, \quad -\infty < \alpha_i < \infty, \quad \sigma^2 > 0$$

against

$$(5.1) \quad H_1 : \beta_1 \neq \beta_2, \quad -\infty < \alpha_i < \infty, \quad \sigma^2 > 0 \quad i = 1, 2.$$

Assume that under both H_0 and H_1 , α_i has a vague prior and that under

$$H_0 : \beta_0 \sim N(\mu, \gamma\sigma^2)$$

and

$$H_1 : \beta_i \sim N(\mu, \gamma\sigma^2), \quad i = 1, 2.$$

Since μ and σ^2 are interchangeable between hypotheses, assume that $p(\mu, \sigma^2) \propto \frac{1}{\sigma^2}$. The numerator of the Bayes factor is then given by

$$(5.2) \quad I_N = (2\pi)^{-\left(\frac{N-3}{2}\right)} (n_1 n_2)^{-\frac{1}{2}} \Gamma\left[\frac{N-3}{2}\right] (SX_1^2 + SX_2^2)^{-\frac{1}{2}} \\ \times \left[\frac{2}{SY_1^2 + SY_2^2 - \frac{(SXY_1 + SXY_2)^2}{SX_1^2 + SX_2^2}} \right]^{\left(\frac{N-3}{2}\right)}$$

where

$$SX_i^2 = \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)^2$$

$$SXY_i = \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)(y_{ij} - \bar{y}_i)$$

$$N = n_1 + n_2$$

The denominator in turn is given by

$$(5.3) \quad I_D = (2\pi)^{-N} \Gamma\left[\frac{N-3}{2}\right] (n_1 n_2)^{\frac{1}{2}} (SX_1^2 + SX_2^2 + 2\gamma SX_1^2 SX_2^2)^{-\frac{1}{2}} \\ \times \left[\frac{2}{SY_1^2 + SY_2^2 - \left[\frac{(SXY_1)^2 (1+2\gamma SX_2^2) + (SXY_2)^2 (1+2\gamma SX_1^2) + 2SXY_1 SXY_2}{SX_1^2 + SX_2^2 + 2\gamma SX_1^2 SX_2^2} \right]} \right]^{\frac{(N-3)}{2}}$$

From (5.2) and (5.3) the Bayes factor follows as

$$(5.4) \quad BF = \left[\frac{SX_1^2 + SX_2^2}{SX_1^2 + SX_2^2 + 2\gamma SX_1^2 SX_2^2} \right]^{\frac{N-4}{2}} \\ \times \left[1 + \frac{2\gamma \{SX_1^2 SX_2^2 (SY_1^2 + SY_2^2) - (SXY_1)^2 SX_2^2 - (SXY_2)^2 SX_1^2\}}{(SY_1^2 + SY_2^2)(SX_1^2 + SX_2^2) - (SXY_1 + SXY_2)^2} \right]^{\frac{N-3}{2}}$$

If one writes

$$F = \frac{(SXY_1 SX_2^2 - SXY_2 SX_1^2)^2 (N-4)}{(SX_1^2 + SX_2^2) \{SX_1^2 SX_2^2 (SY_1^2 + SY_2^2) - (SXY_1)^2 SX_2^2 - (SXY_2)^2 SX_1^2\}}$$

which is the classical test statistic for the hypotheses as given by (5.1) with 1 and $N-4$ degrees of freedom, it then follows that the Bayes factor can be written as

$$(5.5) \quad BF = \left[\frac{SX_1^2 + SX_2^2}{SX_1^2 + SX_2^2 + 2\gamma SX_1^2 SX_2^2} \right]^{\frac{N-4}{2}} \\ \times \left[1 + \frac{2\gamma (SXY_1 SX_2^2 - SXY_2 SX_1^2)^2 (N-4)}{F(SX_1^2 + SX_2^2) \{ (SY_1^2 + SY_2^2)(SX_1^2 + SX_2^2) - (SXY_1 + SXY_2)^2 \}} \right]^{\frac{N-3}{2}}$$

It is seen that the posterior probability is a function of the data not only through F . Figure 5.2 shows the posterior probability λ as a function of γ . It can be seen from the graph that when $\gamma = 0$, $\lambda = \pi_0 = 0.5$, it is the case where the data is ignored completely whereas $\lambda \rightarrow 1$ as $\gamma \rightarrow \infty$.

Of interest here is the lower bound on the posterior probability. The data used is such that the resulting $F_{1, N-4} = 8.8131$ is just significant at the 2.5% level of significance and it can be seen here that the lower bound on λ

is 0,177. This again shows the readiness of a classical test to reject H_0 even when the data has only mild evidence against it.

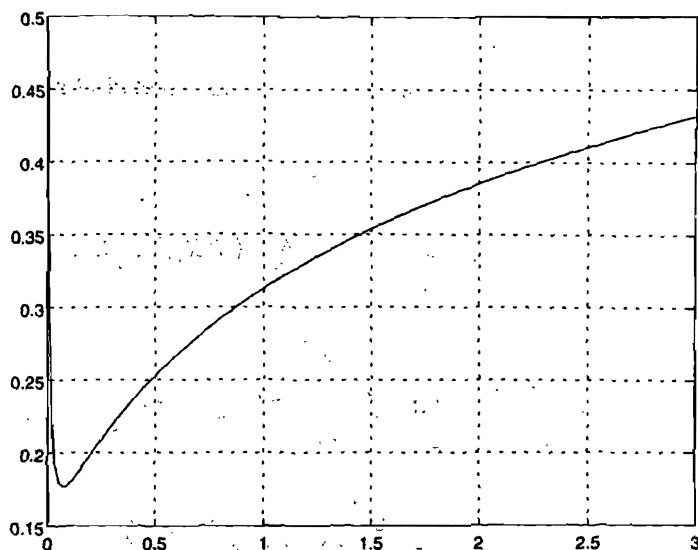


FIGURE 5.1: Posterior probability of H_0 as a function of γ ,
for $N = 10$, $F = 8,8131$

6. TESTING FOR A SUBVECTOR EQUAL TO ZERO VECTOR

This section deals with hypotheses relating to a subvector of $\underline{\beta}$. Given the regression model $\underline{y} = \alpha \underline{e} + X \underline{\beta} + \underline{\varepsilon}$ with assumptions as in section 3, rewrite the model as

$$(6.1) \quad \underline{y} = \alpha \underline{e} + X_1 \underline{\beta}_1 + X_2 \underline{\beta}_2 + \underline{\varepsilon}$$

where $X = (X_1 \ X_2)$ and $X_1 : n \times k_1$ and $X_2 : n \times k_2$ form a partition of the matrix X . The hypotheses of interest are:

$$(6.2) \quad H_A : \underline{\beta}_2 = \underline{0}, \quad -\infty < \alpha, \beta_i < \infty, \quad 0 < \sigma^2 < \infty, \quad i = 1, \dots, k_1$$

against

$$H_B: \beta_2 \neq 0, \quad -\infty < \alpha, \beta_i < \infty, \quad 0 < \sigma^2 < \infty, \quad i = 1, \dots, k.$$

Two approaches which differ in the prior distributions of the unknown parameters will be considered.

6.1 In the first approach consider the following reparameterization which simplifies the computation:

$$(6.1.1) \quad y = \alpha \varepsilon + X_1 \eta + V \beta_2 + \varepsilon$$

where

$$V = [I - X_1'(X_1'X_1)^{-1}X_1']X_2' \quad \text{and} \quad \eta = \beta_1 + (X_1'X_1)^{-1}X_1'X_2\beta_2.$$

Under model 6.1

$$p(y | \alpha, \sigma^2, \eta, H_A) \propto \left[\frac{1}{\sigma^2} \right]^{\frac{n}{2}} \exp \{ -[\nu_A s_A^2 + n(\alpha - \bar{y})^2 + (\eta - \hat{\eta})' X_1' X_1 (\eta - \hat{\eta})] / 2\sigma^2 \}$$

and

$$\begin{aligned} p(y | \alpha, \sigma^2, \eta, \beta_2, H_B) &\propto \left[\frac{1}{\sigma^2} \right]^{\frac{n}{2}} \exp \{ -[\nu_B s_B^2 + n(\alpha - \bar{y})^2] / 2\sigma^2 \} \\ &\times \exp \{ -[(\eta - \hat{\eta})' X_1' X_1 (\eta - \hat{\eta}) + (\beta_2 - \hat{\beta}_2)' V' V (\beta_2 - \hat{\beta}_2)] / 2\sigma^2 \} \end{aligned}$$

with

$$\hat{\eta} = (X_1' X_1)^{-1} X_1' y, \quad \nu_A s_A^2 = (y - \bar{y} \varepsilon - X_1 \hat{\eta})' (y - \bar{y} \varepsilon - X_1 \hat{\eta}),$$

$$\nu_A = n - k_1 - 1, \quad \hat{\beta}_2 = (V' V)^{-1} V' y, \quad \nu_B = n - k_1 - k_2 - 1$$

and

$$\nu_B s_B^2 = (y - \bar{y} \varepsilon - X_1 \hat{\eta} - V \hat{\beta}_2)' (y - \bar{y} \varepsilon - X_1 \hat{\eta} - V \hat{\beta}_2).$$

Using the modified version of the priors given by Zellner and Siow (1980), define the prior under H_A as follows:

$$p(\alpha, \sigma^2, \eta | H_A) \propto |X_1' X_1|^{1/2} / \sigma^2 \quad -\infty < \alpha, \eta_i < \infty, i = 1, \dots, k_1 \quad 0 < \sigma^2 < \infty$$

while under H_B the prior pdf is

$$(6.1.2) \quad p(\alpha, \sigma^2, \eta, \beta_2 | H_B) \propto |X_1' X_1|^{1/2} h(\beta_2 | \gamma \sigma^2) / \sigma^2 \quad -\infty < \alpha, \eta_i < \infty \quad 0 < \sigma^2 < \infty$$

with

$$h(\beta_2 | \gamma \sigma^2) = c_B |V' V / n \gamma \sigma^2|^{1/2} / (1 + \beta_2' V' V \beta_2 / n \gamma \sigma^2)^{(k_2+1)/2} \quad -\infty < \beta_i < \infty \\ i = 1, \dots, k_2$$

where

$$c_B = \Gamma\left[\frac{k_2 + 1}{2}\right] / \pi^{(k_2+1)/2}$$

The above form of prior distributions was suggested by Zellner and Siow (1980) with $\gamma = 1$.

The numerator of the Bayes factor is

$$(6.1.3) \quad I_N = (2\pi)^{\frac{k_1+1}{2}} \left[\frac{1}{n}\right]^{\frac{1}{2}} \Gamma\left[\frac{n - k_1 - 1}{2}\right] \left[\frac{2}{\nu_A s_A^2}\right]^{\frac{n - k_1 - 1}{2}}$$

The denominator is given by

$$I_D = (2\pi)^{\frac{k_1+1}{2}} \left[\frac{1}{n}\right]^{\frac{1}{2}} c_B \left[\frac{1}{\gamma}\right]^{\frac{k_2}{2}} \int \left|\frac{V' V}{n \sigma^2}\right|^{\frac{1}{2}} / (1 + \beta_2' V' V \beta_2 / n \gamma \sigma^2)^{\frac{k_2+1}{2}} \left[\frac{1}{\sigma^2}\right]^{\frac{n - k_1 - 1}{2} + 1} \\ \times \exp\{-[\nu_B s_B^2 + (\beta_2 - \hat{\beta}_2)' V' V (\beta_2 - \hat{\beta}_2) / 2 \sigma^2]\} d\beta_2 d\sigma^2$$

On integrating approximately with respect to the elements of β_2 following Jeffrey's approach and also Cramer's (1946, p. 353) one obtains

$$I_D = (2\pi)^{\frac{k+1}{2}} n^{-\left(\frac{k_2+1}{2}\right)} \gamma^{\frac{k_2}{2}} c_B \int \left[\frac{1}{\sigma^2} \right]^{\frac{n-k_1-1}{2}} \\ \times \left[1 + \frac{\hat{\beta}_2' V' V \hat{\beta}_2}{n \gamma \sigma^2} \right]^{-\left(\frac{k_2+1}{2}\right)} \exp \left\{ -\frac{\nu_B s_B^2}{2 \sigma^2} \right\} d\sigma^2.$$

From Zellner and Siow (1980, p. 602) follows

$$(6.1.4) \quad I_D = c_B (2\pi)^{\frac{k+1}{2}} n^{-\left(\frac{k_2+1}{2}\right)} \gamma^{\frac{k_2}{2}} \Gamma \left[\frac{\nu_A}{2} \right] \left[\frac{2}{\nu_B s_B^2} \right]^{\frac{\nu_A}{2}} \\ \times \left[1 + \frac{\hat{\beta}_2' V' V \hat{\beta}_2}{\gamma \nu_B s_B^2} \right]^{-\left(\frac{k_2+1}{2}\right)}$$

so that

$$(6.1.5) \quad BF = \frac{1}{c_B} \left[\frac{\nu_B \gamma}{2\pi} \right]^{\frac{k_2}{2}} \left[\frac{\nu_B s_B^2}{\nu_A s_A^2} \right]^{\frac{\nu_A}{2}} \left[1 + \frac{\hat{\beta}_2' V' V \hat{\beta}_2}{\gamma \nu_B s_B^2} \right]^{\frac{k_2+1}{2}}$$

where $\frac{n}{2}$ has been substituted by $\frac{\nu_B}{2}$, which is consistent with the approximation in the integration.

If one writes $F_{k_2, \nu_B} = \frac{\hat{\beta}_2' V' V \hat{\beta}_2}{k_2 s_B^2}$ which is the usual F -statistic for testing the hypotheses in (6.1), then the Bayes factor can be written as

$$(6.1.6) \quad BF = \frac{1}{c_B} \left[\frac{\nu_B \gamma}{2\pi} \right]^{\frac{k_2}{2}} \left[1 + \frac{k_2 F}{\nu_B} \right]^{-\frac{\nu_A}{2}} \left[1 + \frac{k_2 F}{\gamma \nu_B} \right]^{\left(\frac{k_2+1}{2}\right)}$$

The Bayes factor is a function of the data only through F , and the unknown gamma.

It is possible to establish the lower bound on the Bayes factor and hence the posterior probability of H_0 with respect to γ . This lower bound occurs at

$$(6.1.7) \quad \hat{\gamma} = \frac{\hat{\theta}'_2 V' V \hat{\theta}_2}{k_2 \nu_B s_B^2},$$

and this gives the minimum Bayes factor as

$$-BF^* = \pi^{\frac{1}{2}} \left[\frac{F}{2} \right]^{\frac{k_2}{2}} \left[\Gamma \left[\frac{k+1}{2} \right] \right]^{-1} \left[1 + \frac{k_2 F}{\nu_B} \right]^{-\frac{\nu_A}{2}} (1 + k_2)^{\frac{(k_2+1)}{2}}.$$

Also, for fixed F ,

$$(6.1.8) \quad \lim_{n \rightarrow \infty} BF^* = BF^{**} = \pi^{\frac{1}{2}} \left[\frac{F}{2} \right]^{\frac{k_2}{2}} \left[\Gamma \left[\frac{k+1}{2} \right] \right]^{-1} (1 + k_2)^{\frac{(k_2+1)}{2}} e^{-\frac{k_2 F}{2}}$$

Table 6.1 gives lower bounds on the posterior probability for H_0 as well as the posterior probabilities for H_0 with $\gamma = 1$, which is the approach by Zellner and Siow, for selected values of n and k with F corresponding to the critical value at the 5% level of significance. Note that for $\gamma = 1$ one must have $n - k_1 - k_2 - 1 > 1$ for the Bayes factor to be a function of the data.

The first row for each of the combinations of n , k and k_2 gives the lower bound on the posterior probability whereas the second row gives the posterior probability for $\gamma = 1$.

TABLE 6.1: Lower bounds on posterior probabilities of H_0 and posterior probabilities of H_0 for $\gamma = 1$

k_2 k	n	5	10	18	32	∞
1 3		0,1639	0,3524	0,3896	0,4040	0,4186
			0,3524	0,4252	0,5017	1
3 5			0,2490	0,3718	0,4162	0,4586
			0,2571	0,4526	0,6750	1
3 9				0,3352	0,4089	0,4586
				0,3567	0,6268	1

6.2 In the second approach assume that under both H_A and H_B , α and β_1 have vague priors with $p(\sigma^2) \propto \frac{1}{\sigma^2}$, and that under H_B , $\beta_2 \sim N(0, \gamma\sigma^2 I)$.

The numerator of the Bayes factor is

$$(6.2.1) \quad I_N = \left[\frac{1}{n} \right]^{\frac{1}{2}} \left[\frac{1}{2\pi} \right]^{\frac{n-k_1-1}{2}} |X_1' X_1|^{-\frac{1}{2}} \left[\frac{2}{\nu_3 s_3^2} \right]^{\frac{n-k_1-1}{2}} \Gamma \left[\frac{n-k_1-1}{2} \right]$$

where

$$\hat{\beta}_1 = (X_1' X_1)^{-1} X_1' y, \quad \nu_3 = n - k_1 - 1$$

and

$$\nu_3 s_3^2 = (y - \bar{y}e - X_1 \hat{\beta}_1)' (y - \bar{y}e - X_1 \hat{\beta}_1).$$

The denominator is

$$(6.2.2) I_D = \left[\frac{1}{n} \right]^{\frac{1}{2}} \left[\frac{1}{2\pi} \right]^{\frac{n-k_1-1}{2}} |X_1' X_1|^{-\frac{1}{2}} |\gamma X_2' X_2 + I|^{-\frac{1}{2}} \Gamma \left[\frac{n-k_1-1}{2} \right] \\ \times \left[\frac{\nu_4 s_4^2 - \gamma \hat{\beta}_2' X_2' X_2 (\gamma X_2' X_2 + I)^{-1} X_2' X_2 \hat{\beta}_2 + \hat{\beta}_2' X_2' X_2 \hat{\beta}_2}{2} \right]^{\frac{n-k_1-1}{2}}$$

where

$$\nu_4 = n - k_1 - k_2 - 1, \quad \hat{\beta}_2 = (X_2' X_2)^{-1} X_2' y,$$

and

$$\nu_4 s_4^2 = (y - \bar{y}e - X_1 \hat{\beta}_1 - X_2 \hat{\beta}_2)' (y - \bar{y}e - X_1 \hat{\beta}_1 - X_2 \hat{\beta}_2)$$

so that

$$(6.2.3) \quad BF = |X_1' X_2 + I|^{\frac{1}{2}} \left[\frac{\nu_4 s_4^2}{\nu_3 s_3^2} \right]^{\frac{n-k_1-1}{2}} \\ \times \left[1 - \frac{\gamma \hat{\beta}_2' X_2' X_2 (\gamma X_2' X_2 + I)^{-1} X_2' X_2 \hat{\beta}_2 - \hat{\beta}_2' X_2' X_2 \hat{\beta}_2}{\nu_4 s_4^2} \right]^{\frac{n-k_1-1}{2}}$$

7. MODEL SELECTION

In this section it is of interest to select the best regression model from different models. Suppose one considers k independent variables and it is intended to select a subset of the k variables which can be considered the best to predict the dependent variable Y .

Let H_{rt} denote the hypothesis that the regression model has r independent variables and t being a particular combination of these r variables, where $r = 0, 1, \dots, k$ and $t = 1, \dots, \binom{k}{r}$. Then the model under H_{rt} can be written as

$$(7.1) \quad Y = \alpha \underline{e} + X_{rt} \underline{\beta}_{rt} + \varepsilon$$

where $\underline{e}$ is an $n \times 1$ vector of ones, α a scalar, $\underline{\beta}_{rt}$ an $r \times 1$ vector of unknown regression coefficients and X_{rt} is an $n \times r$ matrix of rank r . It follows that the number of all possible models is 2^k .

Further assume that

1. the variables in X are measured in terms of deviations from their respective sample means;
2. elements of ε are independently drawn from a normal population with mean zero and unknown finite variance σ^2 .

Two Bayes approaches differing in the prior distributions of the unknown parameters, which can be used to test the 2^k hypotheses will be considered.

7.1 The likelihood functions under the different hypotheses are given by

$$(7.1.1) \quad p(\underline{y} | \alpha, \sigma^2, H_{01}) = \left[\frac{1}{2\pi\sigma^2} \right]^{\frac{n}{2}} \exp[-(\underline{y} - \alpha \underline{e})' (\underline{y} - \alpha \underline{e}) / 2\sigma^2]$$

and

$$(7.2.1) \quad p(\underline{y} | \alpha, \underline{\beta}_{rt}, \sigma^2, H_{rt}) = \left[\frac{1}{2\pi\sigma^2} \right]^{\frac{n}{2}} \exp[-(\underline{y} - \alpha \underline{e} - X_{rt} \underline{\beta}_{rt})' (\underline{y} - \alpha \underline{e} - X_{rt} \underline{\beta}_{rt}) / 2\sigma^2].$$

Suppose further that under H_{01} $p(\alpha, \sigma^2 | H_{01}) \propto \frac{1}{\sigma^2}$ and under H_{rt}

$$p(\alpha, \underline{\beta}_{rt} | \sigma^2) \propto f(\underline{\beta}_{rt} | \sigma^2) / \sigma^2$$

with

$$(7.1.3) \quad f(\underline{\beta}_{rt} | \sigma^2) = c |X'_{rt} X_{rt} / n\gamma\sigma^2|^{\frac{1}{2}} / \left[\frac{\underline{\beta}'_{rt} X'_{rt} X_{rt} \underline{\beta}_{rt}}{n\gamma\sigma^2} \right]^{\frac{r+1}{2}},$$

where

$$c = \Gamma\left[\frac{r+1}{2}\right] / \pi^{\left(\frac{r+1}{2}\right)}$$

and γ is an unknown positive real number. The above follows from Zellner and Siow (1980) apart from the inclusion of γ .

Let

$$A_{01} = \int \int p(y | \alpha, \sigma^2, H_{01}) d\alpha d\sigma^2,$$

$$A_{rt} = \int \dots \int p(y | \alpha, \beta_{rt}, \sigma^2, H_{rt}) p(\alpha, \beta_{rt}, \sigma^2 | H_{rt}) d\alpha d\beta_{rt} d\sigma^2 \quad r=1, \dots, k$$

so that

$$(7.1.5) \quad p(H_{sp} | x) = \frac{A_{sp}}{\sum_{r=0}^k \sum_{t=1}^{q_r} A_{rt}}$$

under the assumption of uniform prior probabilities for all models, and where $q_r = \binom{k}{r}$ is the number of possible combinations of the r independent variables.

It follows from (7.1.1) to (7.1.4) that

$$A_{01} = \left[\frac{1}{2\pi} \right]^{\frac{n-1}{2}} \left[\frac{1}{n} \right]^{\frac{1}{2}} \left[\frac{2}{\nu_0 s_0^2} \right]^{\frac{\nu_0}{2}} \Gamma \left[\frac{\nu_0}{2} \right],$$

and

$$A_{rt} = c \left[\frac{1}{2\pi} \right]^{\frac{n-1}{2}} \left[\frac{1}{n} \right]^{\frac{1}{2}} \left[\frac{1}{\gamma} \right]^{\frac{r}{2}} \left[\frac{2\pi}{\nu_r s_r^2} \right]^{\frac{\nu_0}{2}} \left[1 + \frac{\hat{\beta}'_{rt} X'_{rt} X_{rt} \hat{\beta}_{rt}}{\gamma \nu_r s_r^2} \right]^{-\left(\frac{r+1}{2}\right)} \quad r = 1, \dots, k$$

where

$$(7.1.6) \quad \nu_0 = n - 1, \quad \nu_r = n - r - 1, \quad \hat{\beta}_{rt} = (X'_{rt} X_{rt})^{-1} X'_{rt} y, \\ \nu_0 s_0^2 = \Sigma (y_i - \bar{y})^2 \quad \text{and} \quad \nu_r s_r^2 = (y - \bar{y} \mathbf{1} - X_{rt} \hat{\beta}_{rt})' (y - \bar{y} \mathbf{1} - X_{rt} \hat{\beta}_{rt}).$$

From (7.1.5) and (7.1.6) the posterior probability of any of the 2^k hypotheses can be computed.

An application of this to real data will be given at the end of section 7.2.

7.2 As an alternative approach assume that under H_{01} and all H_{rt} , $\alpha \sim N(m, \gamma_1 \sigma^2)$, $h(m, \sigma^2) \propto \frac{1}{\sigma^2}$ and under H_{rt} , $\beta_{rt} \sim N(0, \gamma_2 \sigma^2 I)$.

It then follows that

$$A_{01}^* = n^{-\frac{1}{2}} (2\pi)^{-\left(\frac{n-1}{2}\right)} \Gamma\left(\frac{n-1}{2}\right) \left[\frac{2}{\nu_0 s_0^2} \right]^{\frac{n-1}{2}},$$

and

$$A_{rt}^* = n^{-\frac{1}{2}} (\pi)^{-\left(\frac{n-1}{2}\right)} \Gamma\left(\frac{n-1}{2}\right) [\gamma_2 X'_{rt} X_{rt} + I_r]^{-\frac{1}{2}} \\ \times (\nu_r s_r^2 - \gamma_2 \hat{\beta}'_{rt} X'_{rt} X_{rt} (\gamma_2 X'_{rt} X_{rt} + I_r)^{-1} X'_{rt} X_{rt} \hat{\beta}_{rt} - \hat{\beta}'_{rt} X'_{rt} X_{rt} \hat{\beta}_{rt})^{-\frac{\nu_0}{2}} \\ r = 1, \dots, k, \quad t = 1, \dots, \binom{k}{r},$$

where A_{01}^* and A_{rt}^* have the same meaning as A_{01} and A_{rt} in (7.1.4), and $\nu_r, s_r^2, \nu_0, s_0^2$ and $\hat{\beta}_{rt}$ are as defined in (7.1.6).

It follows that the posterior probability for H_{sp} is as in (7.1.5).

An application of the two approaches to real data follows.

EXAMPLE 7.1

Consider part of the data in Anderson and Bancroft (1952, p. 205). In this problem 25 samples of tobacco leaf were analysed for organic and inorganic chemical constituents, and regression was used to discover the nature and extent of the relationship of certain of these constituents. In the application use will be made of the following part of the data:

Dependent variable:	Y = Percent nicotine
Independent variables:	Z_1 = Percentage of nitrogen
	Z_2 = Percentage of chlorine
	Z_3 = Percentage of potassium
	Z_4 = Percentage of magnesium

Table 7.1 gives the observed values of the above variables for the 25 samples of tobacco leaf (for complete data see Anderson and Bancroft (1952) or Van der Merwe, Groenewald, De Waal and Van der Merwe (1983)).

TABLE 7.1: Tobacco data

Y	Z_1	Z_2	Z_3	Z_4
1,38	2,02	2,90	2,17	0,91
2,64	2,62	2,78	1,72	1,25
1,56	2,08	2,68	2,40	0,82
2,22	2,20	3,17	2,06	0,97
2,85	2,38	2,52	2,18	1,12
1,24	2,03	2,56	2,57	0,82
2,86	2,87	2,67	2,64	1,06
2,18	1,88	2,58	2,22	1,01
1,65	1,93	2,26	2,15	0,92
3,28	2,57	1,74	1,64	1,22
1,56	1,95	2,15	2,48	0,81
2,00	2,03	2,00	2,38	0,98
2,88	2,50	2,07	2,32	2,04
1,36	1,72	2,24	2,25	0,78
2,66	2,53	1,74	2,64	0,93
2,43	1,90	1,46	1,97	0,90
2,42	2,18	0,74	2,46	0,86
2,16	2,16	2,84	2,36	0,95
2,12	2,14	3,30	2,04	1,06
1,87	1,98	2,90	2,16	0,84
2,10	1,89	2,82	2,04	1,02
2,21	2,07	2,97	2,15	1,04
2,00	2,08	3,14	2,60	0,80
2,26	2,21	2,81	2,18	0,92
2,14	2,00	3,16	2,22	1,07

In the analysis which follows the independent variables have been standardized.

It is noticed that for the approaches derived the posterior probability of a hypothesis depends on the unknown positive constant, γ for the first and γ_2 for the second approach. A study of the behaviour of the posterior probabilities of the hypotheses as functions of these unknown values will be conducted.

A prior probability of $\frac{1}{16}$ was assigned to each hypothesis. In a wide range for γ and γ_2 it was found that the following hypotheses had the largest posterior probabilities for both approaches.

H_{32}	: Z_1, Z_2	and Z_4	as the dependent variables.
H_{41}	: Z_1, Z_2, Z_3	and Z_4	as the dependent variables
H_{23}	: Z_1	and Z_4	as the dependent variables,
H_{31}	: Z_1, Z_2	and Z_3	as the dependent variables,

The rest had negligible posterior support. It is interesting to note that under both approaches H_{32} and H_{31} had the highest and lowest support respectively amongst the four hypotheses.

Figures 7.1 and 7.2 give the graphs of the above hypotheses as a function of γ and γ_2 for approach 1 and 2 respectively. Notice that when γ and γ_2 get larger than two, the two approaches give almost the same posterior probabilities for each of the above hypotheses. The posterior probability of the important hypothesis H_{01} is also negligible and cannot be shown on the graph. Of course, the posterior probability for H_{01} will approach one as γ or γ_2 approaches infinity, the effect of Lindley's paradox. This, however, happens very slowly, and the probabilities in figures 7.1 and 7.2 are quite robust for reasonable values of γ and γ_2 . Notice also that only for approach 2 do the posterior probabilities approach $\frac{1}{16}$ as $\gamma_2 \rightarrow 0$.

It is further noted that the two Bayesian approaches derived agree with the results of Van der Merwe et al. (1983) if applied to the subset considered in this application.

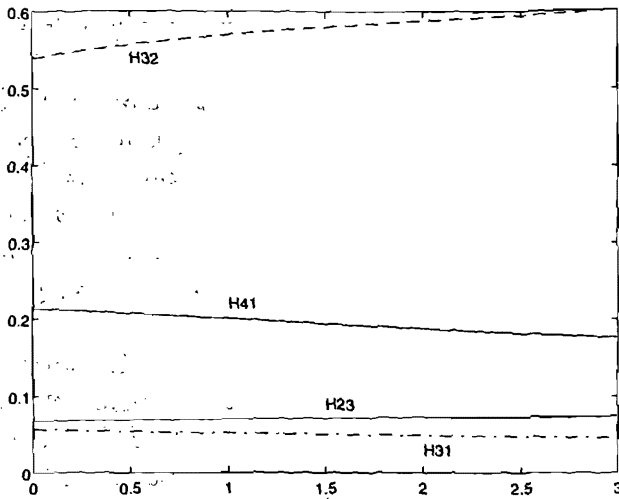


FIGURE 7.1: Posterior probabilities for selected hypotheses using Approach 1

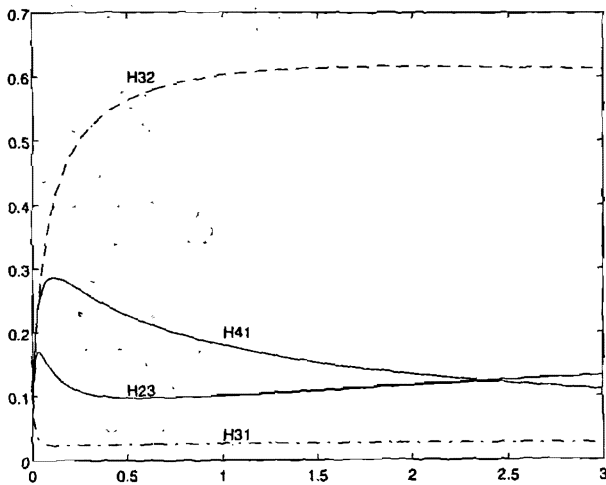


FIGURE 7.2: Posterior probabilities for selected hypotheses using Approach 2

8. CONCLUSION

In this study Bayesian measures of evidence have been investigated in hypothesis testing for regression models.

The results of the study indicate that there is a conflict between these Bayesian evidence as measured by the Bayes factor or the posterior probability of a hypothesis and the sampling-theorists' p -value. This is in line with what has been established by many authors including Jeffreys (1939/1961), Lindley (1957), Edwards, Lindman and Savage (1962), Berger and Sellke (1987), Berger and Delampady (1987), Delampady (1989) and Groenewald and De Waal (1989).

It has been shown that the evidence of the data in support of H_0 is played down by the p -value, so much that it is even lower than the lower bound of the posterior probability of H_0 over a reasonable class of prior distributions. In this study a conjugate class G with a specified mean was used and this is believed to be a reasonable class. Note that in obtaining the lower bound of the posterior probability of H_0 , $a \in G$ which is most favourable to H_1 is used.

There are typical criticisms that can be levelled against the setup as follows in this study. The first one is that posterior probabilities and p -values cannot be compared since the two do not mean the same thing. This has been addressed by several authors, amongst them Berger and Sellke (1987) who stated that:

"At this point, there might be cries of outrage to the effect that $p = 0,05$ was never meant to provide an absolute measure of evidence against H_0 and any such interpretation is erroneous. The trouble with this view is that, like it or not, people do hypothesis testing to obtain evidence as to whether or not the hypotheses are true, and it is hard to fault the majority of non-specialists for assuming that, if $p = 0,05$ (for point null) really means that there is at best very weak evidence against H_0 . Indeed, most non-specialists interpret p precisely as $Pr(H_0|x)$ (see Diamond and Forrester (1983)), which only compounds the problem."

The second is that of the choice of π_i , the prior probability for a hypothesis H_i . In this study $\pi_i = \frac{1}{k}$ where k is the number of hypotheses being considered. This choice is acceptable to many people, and it makes sense to have the same degree of belief on each hypothesis opinion, otherwise one or more hypothesis would be disadvantaged even before data is gathered. Beyer and Sellke (1987) state:

Except for personal decisions (or enlightened through subjective Bayesian hypothesis testing) it will barely be justifiable to choose $\pi_0 < \frac{1}{2}$. Who, after all, would be convinced by the statement, "I conducted a Bayesian test of H_0 , assigning prior probability 0,1 to H_0 , and my conclusion is that H_0 has posterior probability 0,05 and should be rejected".

In saying that, Berger and Sellke (1987) were supporting the choice of $\pi_0 = \frac{1}{2}$ when testing two hypotheses.

The third would be that testing itself is irrelevant; why should one not simply concentrate on confidence sets? The explanation here is that a point like θ_0 may be outside a $(1-\alpha)\%$ confidence set, and yet not be contraindicated by the data and the only way to judge the support of θ_0 from the data is to calculate a Bayes factor (Berger and Delampady, 1987).

The fourth would be that Point Null Hypotheses are unreasonable since it is rare or impossible to encounter hypotheses that are representable as exact point nulls. To the contrary Berger and Delampady (1987) showed that exact point nulls exist and are reasonable and that this point nulls may occur as approximations to fuzzy precise nulls such that the mass π_0 , assigned to $\theta = \theta_0$, is simply the mass corresponding to the precise hypothesis $H_0 : |\theta - \theta_0| \leq \varepsilon$.

From this study and many findings by other authors, use of lower bounds of the posterior probability is recommended. It may not be a perfect measure but it does give a better reflection of the data than the p -value if used with care and responsibility.

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